Faik Kerem Ors
fors@purdue.edu | Personal Page | LinkedIn | GitHub | Google Scholar

Education

Purdue University West Lafayette, IN, US
Ph.D. in Computer Science *Aug. 2022 – May. 2027 (Anticipated)*
***GPA**: 3.90/4.00, **Academic Advisor**: Prof. Elisa Bertino*
***Focus**: Network Security*

Sabancı University Istanbul, Turkey
M.Sc. in Computer Science and Engineering *Sep. 2019 – Jan. 2022*
***GPA**: 3.87/4.00, **Thesis Advisor:** Prof. Albert Levi*
***Thesis**: Data Driven Intrusion Detection for 6LoWPAN Based IoT Systems*

Sabancı University Istanbul, Turkey
B.Sc. in Computer Science and Engineering *Sep. 2014 – June 2019*
Minor in Mathematics
GPA**: 3.86/4.00, **Ranked 3rd

Research Interests

Network security, cellular network security, protocol reverse engineering, IoT security and privacy, systems and software security, machine learning for security.

Research Experience

Research Intern May 2026 – Present
Mitsubishi Electric Research Laboratories (MERL) *Cambridge, MA, USA*

- Conducting research on fault and anomaly detection in large-scale multi-hop IoT networks.
- Designing and implementing fault and attack scenarios in a real IoT testbed to support data collection, evaluation, and model training.
- Developing network anomaly detection techniques leveraging large language models (LLMs) and graph neural networks (GNNs) to detect and analyze network failures and security threats.

Ph.D. Researcher Aug. 2022 – Present
Purdue University - Computer Science *West Lafayette, IN, USA*

- Developing methods to find vulnerabilities in cellular network security, specifically Open Radio Access Networks (O-RAN), using LLMs and formal verification tools.
- Designing and curating large-scale testbeds and datasets for vulnerability analysis and detection in O-RAN.
- Analyzing the security and privacy of communication protocols, particularly unknown or proprietary protocols, using deep learning techniques.
- Developing models and tools to systematically infer protocol structures and behaviors to enhance security analysis, anomaly detection, and formal verification.
- Contributed to the GSMA Open Telco Assets initiative (GSMA and AT&T), including O-RAN grounded synthetic Q&A dataset generation and NDCG-based evaluation of telecom-domain embedding models for retrieval tasks.
- Advisor: Prof. Elisa Bertino

Research Assistant Aug. 2025 – May 2026
Purdue University - Electrical and Computer Engineering *West Lafayette, IN, USA*

- Conducted research on the theoretical foundations of TSD-based pavement evaluation, including deflection theory and pavement response under moving loads.
- Performed literature review, pavement modeling, and index development for TSD-based roadway condition evaluation methods.
- Designed and implemented Rust-based asynchronous clients for sensor data acquisition and device communication over TCP for Traffic Speed Deflectometer (TSD) systems.

- Integrated new sensor systems into a modular, real-time data pipeline using NATS JetStream and Arrow IPC for efficient streaming, serialization, and processing.
- Built frontend applications and APIs to visualize and manage roadway sensor data for the Indiana Department of Transportation (INDOT).
- Presented this work at the *2026 Purdue Road School Transportation Conference and Expo* and the *Joint Transportation Research Program (JTRP).*
- Supervisors: Prof. James V. Krogmeier, Andrew Balmos

CyberPowder Fellow (NSF CyberTraining Award #2417934) Jan. 2026 – May 2026
University of Utah - POWDER Wireless Testbed *Salt Lake City, UT, USA (Remote + On-site)*

- Selected fellow in an NSF-funded wireless networking training and research mentoring program for US-based graduate students.
- Completed weekly research-focused training on wireless communications and hands-on experimentation on POWDER, the world's largest openly-accessible software-defined wireless testbed.
- Participated in a week-long on-site research exploration at the University of Utah, executing an experimental research plan on O-RAN vulnerability validation on the POWDER platform.

Research Assistant – Technical Team Lead June 2025 – Aug. 2025
Purdue University - OATS, IoT4Ag, Agricultural and Biological Engineering *West Lafayette, IN, USA*

- Installed and configured LoRaWAN-based soil and weather sensors, including the deployment of various LoRaWAN gateways as part of the SPRING (Solar-Powered Remote IoT4Ag Network Gateway) inventory.
- Implemented communication interfaces, sensor data decoding logic in Chirpstack, RedPanda Connect flows, TimescaleDB tables, and Grafana dashboards for data pipelining and visualization.
- Built a public-facing dashboard to visualize real-time sensor data and deliver interactive learning content for students, teachers, and the public.
- Collaborated in weekly meetings, shared technical design decisions, and co-developed lecture material, visuals and tutorials that demystify IoT systems for middle-school students.

Research Intern June 2020 – June 2021
Purdue University - GoBoiler Internship Program (Selected Attendee) *West Lafayette, IN, USA*

- Implemented a secure and robust context-based group pairing scheme for heterogeneous IoT devices.
- This work resulted in a publication at *IEEE Symposium on Security and Privacy (S&P 2023).*
- Supervisor: Dr. Z. Berkay Celik

Summer Research Intern June 2018 – Sep. 2018
Technical University of Berlin *Berlin, Germany*

- Implemented deep learning models to optimize the bitrate selection decision on DASH clients.
- Reviewed the literature and delivered an overview presentation on Dynamic Streaming over HTTP (DASH).
- Attended M.Sc. lectures and seminars with the focus on content delivery techniques.
- Supervisors: Dr. Suzan Bayhan and Prof. Abdel-Karim Al-Tamimi

Peer-Reviewed Publications

- **Faik Kerem Ors**, Rumeysa Gorgulu, Andrew D. Balmos, Fabio A. Castiblanco, Joshua K. Bailey, Kristin Korcha, Kwabena Bayity, James V. Krogmeier, and Yaguang Zhang. SPRING Portal: A Multi-Site IoT Data Platform for Classroom-Ready Agricultural Education. In *ASABE AIM 2026*, Indianapolis, IN, July 2026. *(Accepted)*
- **Faik Kerem Ors**, Andrew D. Balmos, Rumeysa Gorgulu, Sofia Gerena Godoy, James V. Krogmeier, and Yaguang Zhang. Agriculture-Focused Introduction to Data Visualization with Grafana: A University-Level Tutorial Module. In *ASABE AIM 2026*, Indianapolis, IN, July 2026. *(Accepted)*
- Rumeysa Gorgulu, **Faik Kerem Ors**, Kristin Korcha, and Yaguang Zhang. From SEED to SPRING: A Sensor-Based Agriculture Module for Middle School Education. In *ASABE AIM 2026*, Indianapolis, IN, July 2026. *(Accepted)*
- Habiba Farrukh*, Muslum Ozgur Ozmen*, **Faik Kerem Ors**, Z. Berkay Celik. One Key to Rule Them All: Secure Group Pairing for Heterogeneous IoT Devices. In *IEEE Security and Privacy (S&P 2023).* – **Cited by 32**
- **Faik Kerem Ors**, and Albert Levi. Data driven intrusion detection for 6LoWPAN based IoT systems. In *Ad Hoc Networks, 143*, pages 103-120, April 2023. – **Cited by 15**

- **Faik Kerem Ors**. Data Driven Intrusion Detection for 6LoWPAN Based IoT Systems. *M.Sc. Thesis*, December 2021.
- **Faik Kerem Ors**, Mustafa Aydin, Aysu Bogatarkan, and Albert Levi. Scalable Wi-Fi Intrusion Detection for IoT Systems. In *11th IFIP International Conference on New Technologies, Mobility and Security (Security Track)*, Paris, France, April 2021. – **Cited by 9**
- **Faik Kerem Ors**, Suveyda Yeniterzi, and Reyyan Yeniterzi. Event Clustering within News Articles, In *Proceedings of the Workshop on Automated Extraction of Socio-political Events from News 2020*, pages 63–68, Marseille, France, May 2020. European Language Resources Association (ELRA). (Proposed system ranked 1st in the shared task). – **Cited by 29**

Conference Presentations

- Pavement Monitoring: Integrating Edge Computing and High-Speed Deflectometry. In *2026 Purdue Road School Transportation Conference and Expo*, West Lafayette, IN, March 2026.
- Scalable Wi-Fi Intrusion Detection for IoT Systems. In *11th IFIP International Conference on New Technologies, Mobility and Security (Security Track)*, Paris, France, April 2021.
- Event Clustering within News Articles, In *Proceedings of the Workshop on Automated Extraction of Socio-political Events from News 2020*, pages 63–68, Marseille, France, May 2020. European Language Resources Association (ELRA).

Poster Presentations

- *Development of a Traffic Speed Deflectometer for Indiana.* (Poster) Joint Transportation Research Program (JTRP), Indiana Government Center North, Indianapolis, IN, March 4, 2026.

Teaching Experience

Invited Lecturer — Fall 2026
ASM 591: Ag Data Visualization & Edge Computing, Purdue University — *West Lafayette, IN, US*
- Delivered a lecture and lab, and assigned coursework on database management systems, SQL, and Grafana for agricultural and weather data visualization.

Teaching Assistant — Aug. 2022 – May 2025
Purdue University — *West Lafayette, IN, US*
- Held office hours, supervised student projects, designed and graded exams and assignments.
- Courses (reverse chronological): Computer Security (CS 426; Spring 2025, Fall 2023, Spring 2023), Cryptography (CS 555; Fall 2024), Information Security (CS 526; Spring 2024), Security Analytics (CS 529; Fall 2022)

Guest Lecturer — Fall 2023
CS 426: Computer Security, Purdue University — *West Lafayette, IN, US*
- Delivered lectures on buffer overflows, return oriented programming, SQL injection, and cross-site scripting.

Teaching Assistant — Feb. 2018 – Jan. 2022
Sabancı University — *Istanbul, Turkey*
- Held office hours, designed and graded assignments, conducted lab sessions and supervised student projects.
- Courses (reverse chronological): Computer Networks (CS 408; 2022, 2021, 2020), Computer and Network Security (CS 432), Machine Learning (CS 412), Advanced Programming (CS 204), Database Systems (CS 306)

Services

- Reviewer in IEEE Transactions on Information Forensics and Security (TIFS) 2026, 2025, 2024
- External Reviewer in NDSS 2026, 2025, 2024
- External Reviewer in IEEE S&P 2026, 2024
- External Reviewer in USENIX Security 2026, 2024
- Reviewer in ITU Journal on Future and Evolving Technologies (ITU J-FET) 2022
- Reviewer in IEEE International Conference on Communications (ICC) 2022
- Reviewer in IEEE Conference on Communications and Network Security (CNS) 2021
- Reviewer in The Computer Journal (Oxford University Press) 2021, 2020

Honors and Awards

- Selected as a CyberPowder Fellow, an NSF-funded wireless networking research program (December 2025).
- Graduate Teaching Award in Recognition of the Teaching Performance during Spring 2025 (November 2025).
- Tuition waiver and Graduate Teaching Assistantship offer by Purdue University for graduate studies (2022 – 2027).
- Full tuition waiver and stipend by Sabancı University for graduate studies (2019 – 2021).
- Dean's High Honor List, Sabancı University (2016 – 2019).
- Recipient of Sakıp Sabancı Encouragement Scholarship, which covers 100% of tuition fee, because of academic excellence (2016 – 2019).

Technical Skills

Programming Languages: Python, C++, C, C#, Java, Rust, SQL
Frameworks and Libraries: PyTorch, Tensorflow, Pandas, NumPy, Scikit-learn, Keras, Flask, Django
Operating Systems: Unix, Linux, macOS, Windows
Technologies: git, MySQL, PostgreSQL, Docker, JUnit, Android Studio, Chirpstack, Redpanda Connect, Grafana
Tools: Wireshark, Metasploit, Hashcat, Burp Suite, Nmap, SQLmap, Wfuzz, IDA, Binwalk, The Harvester, Dirbuster

Additional Work Experience

R&D Engineer — Feb. 2019 – Aug. 2019
PRODAFT — *Istanbul, Turkey*

- Implemented a machine learning based phishing detection system from scratch.
- Developed RESTful microservices to be integrated into the threat intelligence ecosystem of the company.

Security Research Intern — July 2017 – Sep. 2017
PRODAFT — *Istanbul, Turkey*

- Worked on penetration testing and developed penetration testing tools in Python.

Outreach

- Developed and led a hands-on session on cyber-physical systems security for high school students at the 4-H Academy @ Purdue (August 2024), featuring interactive demonstrations on the security of robotic vehicles, AR/VR devices, autonomous vehicles, and industrial control systems.